

Name: _____

CIS 351 Sample CL2 Problem

1 September 2025

CL2: Read Circuit Descriptions

- (a) Complete the truth table below so that it describes the output of the Boolean expression.

x	y	z	$(\overline{xy})(\overline{z})$
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

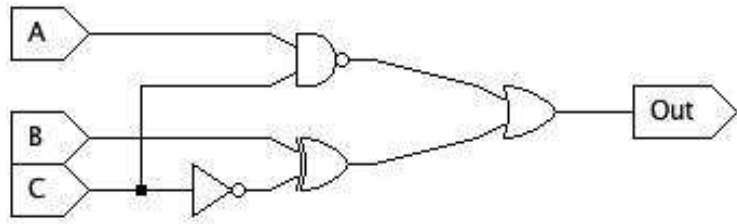
- (b) Complete the truth table below so that it describes the output of the Boolean expression.

x	y	z	$xy + \bar{z}$
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

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(c) Complete the truth table below to show the output of the circuit for each input:

A	B	C	Out
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	



CIS 351 Sample CL2 Problem Solutions

Thu 24th Sept, 2026

CL2: Read Circuit Descriptions

- (a) Complete the truth table below so that it describes the output of the Boolean expression.

x	y	z	$(\overline{xy})(\overline{z})$
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	0

- (b) Complete the truth table below so that it describes the output of the Boolean expression.

x	y	z	$xy + \overline{z}$
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

(c) Complete the truth table below to show the output of the circuit for each input:

A	B	C	Out
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

